

MD ANISUR RAHMAN CHOWDHURY

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RESEARCH SUMMARY

Distributed systems researcher and cloud, systems, and network solution engineer with 7+ years of enterprise infrastructure experience spanning ISP operations, corporate IT, and cloud-native environments. My research centers on building security and reliability directly into the execution and orchestration layers of cloud-native and serverless systems, drawing on distributed system design, large-scale cloud infrastructure, and machine learning to produce deployable, low-latency defenses for data-centric platforms. Author or co-author of 25 peer-reviewed papers (15 as first author, 16 already published in IEEE venues), cited 105+ times with an h-index of 5 on Google Scholar, and an active reviewer for two Scopus-indexed journals. I hold an M.S. in Computer and Information Science, Information Technology specialization (GPA 3.95/4.00), from Gannon University, and am CCNA, JNCIA, and MTCNA certified. I am looking for a funded Ph.D. position to extend this work toward provably secure, self-adapting infrastructure for cloud and edge systems.

RESEARCH INTERESTS

- Serverless and cloud-native security architecture
- Zero Trust systems and distributed policy enforcement
- Machine learning for intrusion, anomaly, and threat detection
- Federated and privacy-preserving learning across untrusted tenants
- Multi-agent LLM orchestration and self-correcting reasoning
- Network and distributed systems design at enterprise scale

EDUCATION

M.S., Computer and Information Science (Information Technology) — Gannon University, Erie, PA *May 2026*

GPA: 3.95 / 4.00. Graduate Research Assistant, CIS Department.

B.S., Electronics & Telecommunication Engineering — University of Science & Technology Chittagong, Bangladesh *Jul 2017*

SELECTED RESEARCH PROJECTS

Serverless Intelligent Firewall (SIF)

IEEE CSCloud 2025; IEEE SmartCloud 2026

An event-driven cloud security platform that watches traffic and access patterns inside AWS Lambda, WAF, IAM, and S3, and enforces Zero Trust policy automatically rather than through static rule sets. Three classifiers — LSTM, BiGRU, and XGBoost — feed a single detection pipeline that reached 98% multi-class accuracy and a 0.990 ROC-AUC across five attack categories on the CICIDS2017 benchmark, with a 135 ms average response latency. The follow-up paper, published at IEEE SmartCloud 2026, extends the design to cross-cloud adaptation.

FedThreat-X

IEEE QPAIN 2026

A federated learning framework that lets multiple cloud tenants train a shared threat-detection model without exposing their own network traffic to one another. Built for organizations that see only part of an attack pattern and have no practical way to compare notes without a formal data-sharing agreement.

PentAI Pro

IEEE paper in preparation

A self-hosted platform that runs LLM-driven agents across three isolated virtual machines to carry out a full penetration-testing sequence — reconnaissance, enumeration, vulnerability mapping, exploitation planning — without sending data to any outside service. Scope is checked three separate times before a tool is allowed to run, and every step is signed and chained into a SHA-256 tamper-evident audit log. Built on a local Ollama deployment (qwen2.5:14b for planning, llama3.2:3b for fast classification), with a 108-test suite passing in under 44 seconds.

Distributed AI — Multi-Agent Ensemble and Self-Correcting Reasoning

IEEE conference (in review / in prep)

Two related systems built around the same question: how a set of local language models can check and improve on each other's work instead of trusting a single pass. The ensemble system coordinates four independently run inference agents through five aggregation strategies, including majority voting and a two-round debate format, validated with paired t-tests and Wilcoxon signed-rank tests on MMLU, GSM8K, and TruthfulQA. The self-correcting framework separates the task of producing an answer from the task of reviewing it, then reconciles both in a third stage — lifting GSM8K accuracy from 31.2% to 49.9%, a 60% relative gain, with a 78% error-correction rate.

Open-Source WAF Implementation (ModSecurity)

IEEE conference (in preparation)

“Open-Source Web Application Firewall Implementation Using ModSecurity Engine: A Performance and Security Evaluation Framework.” Tests how different ModSecurity rule-set configurations trade off detection coverage against request latency and throughput under load. The goal is a reproducible benchmark that other teams can run against their own rule sets, rather than a one-off comparison — extending the policy-enforcement work behind the Serverless Intelligent Firewall into an open-source, self-hostable setting. Collaboration with Dr. Samuel Tweneboah-Koduah, Campbellsville University.

PUBLICATIONS

25 peer-reviewed papers — 16 published, 1 accepted, 1 under review, 7 in preparation. First author on 15 of 25. 105+ citations, h-index 5 (Google Scholar).

First author, published or accepted

1. Towards a Serverless Intelligent Firewall: AI-Driven Security, and Zero-Trust Architectures — *IEEE CSCloud 2025, Nov 2025*
Chowdhury, Dang, Bazan-Antequera, Khan, Karim, Manakkadu
2. Auction-Based Dynamic Resource Allocation for Optimized Edge Computing in Distributed Networks — *IEEE CSITSS 2025, Nov 2025*
Chowdhury et al.
3. AI and Cloud Computing in Business Systems: A Hybrid Model for Enhancing Enterprise Resource Planning — *IEEE CSITSS 2025, Nov 2025*
Chowdhury et al.
4. Deepfake Detection in MIS: Leveraging DenseNet and Multi-Scale Information for Enhanced Digital Forensics — *IEEE COMPAS 2025, Oct 2025*
Chowdhury et al.
5. FedThreat-X: A Privacy-Preserving Federated Threat Intelligence Framework for Multi-Cloud Cybersecurity — *IEEE QPAIN 2026, Apr 2026*
Chowdhury et al.
6. Towards a Serverless Intelligent Firewall: Integrating Cross-Cloud Adaptation, AI-Driven Security, and Zero-Trust Architectures — *IEEE SmartCloud 2026, May 2026*
Chowdhury, Wang
7. Teaching Large Language Models to Think Twice: A Three-Stage Framework for Self-Correcting Mathematical Reasoning — *American CSE AIR-RES/CAC 2026*
Chowdhury, Wang · accepted
8. DataMentor: A Practical Framework for Serverless CSV Intelligence with Interactive Notebook Automation and Deterministic Runtime Repair — *IEEE ISAIA 2026*
Chowdhury, Bazan-Antequera · under review, submission 61
9. Autonomous Self-Learning Serverless Intelligent Firewall: Integrating REST API-Driven Open-Source Threat Intelligence, Multi-Paradigm Machine Learning, and Federated Zero-Trust Architectures — *target: Q1 journal*
Chowdhury, Wang · in preparation
10. A Local Distributed Multi-Agent LLM Ensemble System: Design, Optimization, Reproducibility, and Statistical Evaluation — *target: IEEE conference*
Chowdhury, Wang · in preparation
11. A Hybrid AI/Machine Learning Approach for Real-Time Anomaly Detection in IoT Networks — *target: IEEE conference*
Chowdhury · in preparation
12. Self-Hosted LLM Orchestration for Autonomous Penetration Testing with Three-Layer Scope Enforcement and Tamper-Evident Cryptographic Audit Chains — *target: IEEE conference*
Chowdhury, Bazan-Antequera · in preparation
13. AI-Based Intrusion Detection for IoT Networks — *target: IEEE conference*
Chowdhury, Khan · in preparation
14. Open-Source Web Application Firewall Implementation Using ModSecurity Engine: A Performance and Security Evaluation Framework — *target: IEEE conference*
Chowdhury, Tweneboah-Koduah · in preparation
15. Trust-Based Video Management Framework for Social Multimedia Networks — *target: IEEE conference*
Chowdhury, Tweneboah-Koduah · in preparation

Co-authored, published

16. Enhancing Signature-Based Intrusive Detection System (IDS) for IoT Networks Using Machine Learning Algorithm — *IEEE QPAIN 2025, Jul 2025*
Ahmed, Chowdhury, et al. · 2nd author
17. Cloud-Based CRM Systems Enhanced by AI and Graph Theory: A Hybrid Model for Optimizing Customer Engagement — *IEEE ISAECT 2025, Dec 2025*
Ahmed, Chowdhury, et al. · 2nd author
18. Attention-Enhanced U-Net Models for Breast Cancer Image Analysis: A Comparative Study — *IEEE ECCE 2025, Feb 2025*
Ahmed, Chowdhury, et al. · 2nd author
19. Fed-ZTA-Transformer: A Privacy-Preserving Federated Framework for Continuous Verification in Zero Trust Architectures — *IEEE QPAIN 2026, Apr 2026*
Ahmed, Chowdhury, et al., Bazan-Antequera · 2nd author
20. Detecting Misinformation with Multimodal AI: Leveraging Vision and NLP for Fact-Checking — *IEEE QPAIN 2025, Jul 2025*

Ahmed, et al., Chowdhury · 3rd author

21. A Collaborative Hybrid CNN–LSTM Framework for Real-Time Multichannel Retail Demand Forecasting Using Edge–Cloud Computing — *IEEE QPAIN 2026, Apr 2026*

Aashish KC, Ahmed, Goyal, Chowdhury, et al. · 4th author

22. AI-Enhanced Adaptive Network Security for 6G and Edge Computing — *IEEE QPAIN 2025, Jul 2025*

Ahmed, et al., Chowdhury · 6th author

23. Leveraging Machine Learning and NLP for Adaptive Education Systems: A Personalized Approach for Children — *IEEE QPAIN 2025, Jul 2025*

Ahmed, et al., Chowdhury · 6th author

24. Enhanced Brain Tumor Detection Using Finetuned Transfer Learning Models: Achieving Superior Accuracy with Xception — *IEEE ECCE 2025, Feb 2025*

Ahmed, et al., Chowdhury · 6th author

25. Driving Industry 4.0 with Digital Twins: Enhancing Predictive Maintenance and Operational Performance Through IoT and Machine Learning — *IEEE QPAIN 2025, Jul 2025*

Ahmed, et al., Chowdhury · 7th author

PEER REVIEW & EDITORIAL SERVICE

- Reviewer, Journal of Information Systems Engineering and Management (JISEM), Scopus Q4-indexed — reviewed a submission on AI applications against cyber threats in financial institutions, 2026
- Reviewer, International Journal of Drug Delivery Technology (IJDDT), ISSN 0975-4415, Scopus Q3 (Pharmaceutical Science) — reviewed “A Web Based Honeypot System Framework to Capture and Mitigate Cyber Attacks,” 2026

PROFESSIONAL EXPERIENCE

IT Systems & Operations Engineer — Adel Group Inc., Winnetka, CA *Jul 2026 – Present*

- Administer network and endpoint infrastructure for a nationwide, multi-site retail operation, including firewall policy and access control
- Introduce automation drawn from ongoing cloud-security research to cut manual ticket handling

Graduate Research Assistant — Gannon University, CIS Department, Erie, PA *Aug 2025 – May 2026*

- Designed and built the serverless cloud-security pipeline behind the Serverless Intelligent Firewall research
- Co-authored and presented papers at IEEE CSCloud 2025 and IEEE CSITSS 2025; mentored incoming graduate students

Cybersecurity Lecturer (CPT) — TAAS Services LLC, Remote *Aug 2025 – Dec 2025*

- Built 15+ hands-on lab environments covering perimeter defense, threat analysis, and vulnerability assessment
- Translated MITRE ATT&CK and NIST frameworks into certification-aligned exercises; ran incident-response drills

Senior Executive – Information Technology — S. Alam Group, Chittagong, Bangladesh *Dec 2021 – Jul 2024*

- Ran hybrid Linux/Windows server infrastructure for 500+ users, including mail, web hosting, and DNS/DHCP under SLA
- Built Active Directory identity infrastructure and led LAN/WAN and IP-PBX/VoIP rollouts across multiple sites

Technical Support Engineer – Networking — Link3 Technology Ltd., Bangladesh *Aug 2017 – Dec 2021*

- Managed the solution lifecycle for 100+ enterprise accounts across Cisco, MikroTik, Juniper, Fortinet, and Cyberoam equipment
- Engineered VPN, firewall policy, and multi-link WAN failover architectures for enterprise clients

TECHNICAL SKILLS

Cloud & Architecture: AWS (EC2, VPC, CloudFront, S3, Lambda, WAF), serverless design, event-driven systems, DC/DR planning, CI/CD, Proxmox, VMware ESXi, Hyper-V

Security: Zero Trust design, IDS/IPS, DDoS mitigation, RBAC, SIEM, incident response, OWASP, mTLS, tamper-evident audit logging

Machine Learning: LSTM, BiGRU, XGBoost, CNN, SVM, DenseNet, self-correcting reasoning pipelines, local inference (Ollama: Qwen2.5, Llama3.2, Phi3, Gemma2)

Networking: BGP, OSPF, VLANs, NAT, ACLs, TCP/IP, DNS, TLS/SSL, REST APIs, SD-WAN, CDN

Languages & Frameworks: Python, JavaScript, TypeScript, SQL, Java, Next.js, React, FastAPI, Tailwind CSS, Firebase, Docker, PostgreSQL, GitHub

CERTIFICATIONS

CCNA — Cisco Certified Network Associate (2024) · JNCIA — Juniper Networks Certified Associate (2022) · MTCNA — MikroTik Certified Network Associate (2019)